Constantly Out of Order: A Zero-Query Ring-Permutation Forgery against Constant-Size LRS

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23 October 2026

AI provenance and review status. This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. OpenAI Codex generated and selected this paper's title and the scheme-specific "Constantly Out of Order" wordplay. OpenAI Codex performed AI-assisted target reconstruction, attack-history searching, cryptanalysis, implementation, experiment coordination, and technical writing. A separate AI paper/profile/algebra/security-model review independently rederived the attack and passed its gate. A fresh-context AI reproduction executed the frozen full profile twice and independently validated the evidence. A separately tasked AI adversarial reviewer returned GO with all ten binding qualifications preserved. The final exact-target history recheck and an independent AI publication-package review passed. The result is classified as `VERIFIED_ATTACK`. Independent human verification has not been performed. It is optional, and there is no human-verification requirement.

Abstract

We analyze revision 2 of *Efficient Constant-Size Linkable Ring Signatures for Ad-Hoc Rings via Pairing-Based Set Membership Arguments*, IACR ePrint 2025/1174. Here "zero-query" means zero signing-oracle queries; the construction still makes the ordinary Fiat–Shamir random-oracle queries. The optimized set-membership argument derives its base challenges from the message tag and commitments but omits the ordered ring $\Phi = (\phi_1, \ldots, \phi_n)$. We exploit that omission at the authors' full BN254 evaluation profile, $n = m = 1024$.

Before the aggregate challenge, an attacker pairs 508 honest KGO public-key hashes into 254 unordered pairs and fixes one coefficient per pair as $2^j/(a_j - c_j)$. After the challenge, orienting pair j changes the weighted membership residue by exactly -2^j. Since the BN254 scalar modulus is less than 2^{254}, the 254 orientations span every required field residue. The attacker thereby cancels the sole coefficient whose SRS power is unavailable, constructs the SMA proof from public powers, and computes the later ring-bound KZG opening honestly. A separately generated valid secret outside the ring supplies honest witnesses for ccGro16 and CPlink. The final ring is a permutation of 1,024 distinct honest KGO outputs and does not contain the attacker's key.

A prospectively sealed standard-library implementation executed the exact SMA Equation 15 and KZG Equation 17 polynomial identities at the full paper profile. The deterministic Python paper model, not an end-to-end author implementation, completed in 63.85 seconds using 24,576 KiB peak RSS on one worker; a hostile `python -O` replay completed in 64.27 seconds using 28,220 KiB. Both emitted byte-identical evidence. A separate validator checked all 1,024 BabyJubJub curve equations and MiMC91 hashes plus the pair-swap arithmetic. Fixed-order, one-pair-flip, and changed-message controls rejected. The experiment does not emit concrete ccGro16 or CPlink proofs: it composes their honest-witness relations by correctness. No complete author LRS verifier, outer codec, or known-answer test was located in the published artifact, so we make no byte-level author-verifier claim. No secret-key recovery is claimed.

1 Introduction

Xie, Tu, Au, Fang, Wang, and Jiang propose a constant-size scoped linkable ring signature for ad-hoc rings [1]. Its ordinary verification is a conjunction of three proof systems. An optimized pairing-based

set-membership argument (SMA) is intended to show that a committed public-key hash belongs to the ring; ccGro16 shows correct public-key and link-tag computation; and CPlink ties the committed values together. The paper's evaluated BN254 profile uses BabyJubJub public keys, a two-input 91-round MiMC7 hash, and rings of up to 1,024 members. The paper describes this row as 128-bit and reports a 687-byte algebraic signature. We use "authors' claimed 128-bit row" without independently endorsing that contemporary security estimate.

The attack is a statement-binding failure in the optimized SMA transcript. The aggregation challenge is fixed before the ordered ring is hashed. Its soundness analysis treats every missing-degree relation error as fixed at that time, but the membership error still depends on the ring. An adversary can therefore retain 254 independent pair orientations until after seeing the challenge and use them as a complete binary basis for the BN254 scalar field. This is not a collision search, subset-sum search, powers-of-tau attack, or malformed-key attack.

Contribution. We derive the exact unavailable-degree coefficient in Equation 15, give a deterministic full-profile pair-orientation algorithm, prove that it yields a false-membership signature under the paper semantics, connect the result to the authors' unforgeability game, and report a prospectively sealed full-parameter execution with independently checked controls. We do not claim novelty for the general requirement that Fiat–Shamir challenges bind their complete statement [2], nor for polynomial commitments or missing-power aggregation [3]. The contribution is the concrete attack on this scheme and profile.

Claim boundary. The positive execution is exact for the SMA Equation 15 and KZG Equation 17 algebra. The other two conjuncts use ordinary honest witnesses and are composed by completeness; no concrete ccGro16 or CPlink proof bytes are emitted. The published source is a partial benchmark and reference artifact, not a complete external LRS verifier. Our attack is therefore a paper-level design result, not a wire-format or author-binary conformance claim.

2 Target and frozen profile

The normative target is ePrint 2025/1174, revision 2 dated 24 November 2025 [1]. The retained PDF has SHA-256 digest

`4badb60ab91e825303575acea02ddad0d8bc7bc9fe883aef4dc694323a56d2d6.`

The paper is the full version of an ACM CCS 2025 publication. The frozen row is

`eprint-2025-1174-v2-bn254-babyjubjub-mimc91-n1024-m1024.`

It fixes the BN254 scalar field, the paper's BabyJubJub public-key map, two-input MultiMiMC7 with 91 rounds, $n = m = 1024$, ccGro16, CPlink, and the optimized KZG-backed SMA.

The scalar modulus is

$$q = 21888242871839275222246405745257275088548364400416034343698204186575808495617 < 2^{254}. \tag{1}$$

The ring $\Phi = (\phi_1, \ldots, \phi_m)$ is ordered. Each ϕ_i is the hash of a public key produced through the same public-key mapping. The security game permits the adversary to select an ad-hoc ring from KGO outputs and does not prescribe a canonical ring order before proof commitments.

3 The optimized SMA relation

We work in exponents over $\mathbb{F}_q$. Let X be a formal variable representing the powers-of-tau point and let the outside attacker key hash be $\phi_\star$. For commitment randomness r, r_b, r_s, define

$$A(X) = r + \phi_\star X, \tag{2}$$

$$B(X) = r_b + \sum_{i=1}^{n} b_i X^i, \tag{3}$$

$$S(X) = r_s + \sum_{i=1}^{n} s_i b_i X^{n+1-i}. \tag{4}$$

The attacker forms S in the honest way from $s_i b_i$, even though the chosen b_i are not bits.

For $m = n$, the public polynomials entering Equation 15 are

$$P_h(X) = \sum_{i=1}^{n} \big[\delta_o + (\delta_{eq} t_i - \delta_b) s_i + \delta_\phi \phi_i\big] X^{n+1-i}, \tag{5}$$

$$P_u(X) = \delta_\phi X^n - \delta_{d2} X^{n-1} - \sum_{i=2}^{n} \delta_{d1} u_i X^{n+1-i}, \tag{6}$$

$$P_t(X) = \sum_{i=1}^{n} \delta_{eq} t_i X^i. \tag{7}$$

The required G1 proof exponent is

$$F(X) = B(X)\big[\delta_b S(X) + P_h(X)\big] - A(X)P_u(X) - S(X)P_t(X) - \delta_o X^{n+1}. \tag{8}$$

Its degree is at most $2n$. The public G1 SRS contains every needed power in that range except X^{n+1}. Consequently a prover using only the public SRS can form the proof exactly when the coefficient at degree $n + 1$ is zero.

3.1 The sole unavailable coefficient

Collecting degree $n + 1$ in Equation (8) gives

$$\begin{aligned}[X^{n+1}]F = \delta_b \sum_i s_i b_i^2 + \sum_i b_i \big[\delta_o + (\delta_{eq} t_i - \delta_b) s_i + \delta_\phi \phi_i\big] \\ - \delta_{eq} \sum_i s_i b_i t_i - \delta_\phi \phi_\star - \delta_o.\end{aligned} \tag{9}$$

The two δ_{eq} terms cancel exactly because S in Equation (4) is formed honestly. Thus

$$E = [X^{n+1}]F = \delta_b \sum_i s_i b_i (b_i - 1) + \delta_o \left(\sum_i b_i - 1 \right) + \delta_\phi \left(\sum_i b_i \phi_i - \phi_\star \right). \tag{10}$$

The three summands are respectively the binary, one-hot, and membership errors aggregated by the verifier.

3.2 Where statement binding is lost

The base oracles H_s, H_t, H_u, and H_{agg} absorb the message tag and the relevant commitments, but not the ordered ring Φ. The paper's message-binding remark adds the message tag to these inputs and does not add the ring. Therefore A, B, S, every s_i, and all six aggregation coefficients can be fixed while pair orientations inside Φ remain open.

The outsourced KZG point is sampled later. Its transcript includes the final ring and the commitments to P_h, P_u, and P_t. This later binding is useful for a control, but it cannot retroactively make the membership error in Equation (10) fixed before H_{agg}.

4 Adaptive ring-permutation attack

The adversary generates a valid key $(sk_\star, \phi_\star)$ and obtains a pool of 1,024 distinct honest KGO public-key hashes not containing $\phi_\star$. Choose 254 disjoint unordered pairs

$$(a_j, c_j), \qquad 0 \le j < 254,$$

using 508 pool entries. The other 516 entries occupy zero-coefficient positions. Initially place a_j in the first position of pair j and c_j in the second. Before querying the aggregate oracle, set

$$b_{2j+1} = \frac{2^j}{a_j - c_j}, \qquad b_{2j+2} = 0. \tag{11}$$

All denominators are public and nonzero; the negligible event of equal hashes inside a selected pair is detectable and handled by replacing that pair with another honest KGO output.

Swapping the two entries of pair j changes its contribution to the weighted membership sum by

$$\frac{2^j c_j}{a_j - c_j} - \frac{2^j a_j}{a_j - c_j} = -2^j. \tag{12}$$

After deriving δ, define the already fixed errors

$$E_b = \sum_i s_i b_i (b_i - 1), \qquad E_o = \sum_i b_i - 1, \tag{13}$$

and, provided $\delta_\phi \neq 0$, the target weighted value

$$W = \phi_\star - \frac{\delta_b E_b + \delta_o E_o}{\delta_\phi}. \tag{14}$$

Let

$$B_0 = \sum_{j=0}^{253} \frac{2^j a_j}{a_j - c_j}, \qquad D = (B_0 - W) \bmod q, \qquad 0 \le D < q. \tag{15}$$

By Equation (1), D has a 254-bit expansion

$$D = \sum_{j=0}^{253} \epsilon_j 2^j, qquad \epsilon_j \in \{0, 1\}.$$

Swap pair j exactly when $\epsilon_j = 1$. Equation (12) then gives

$$\sum_i b_i \phi_i = B_0 - \sum_j \epsilon_j 2^j = B_0 - D = W \pmod q. \tag{16}$$

Substitution of Equation (16) into Equation (10) yields $E = 0$ exactly.

4.1 Complete attack chronology

The following order respects every hash dependency in revision 2.

1. Generate the outside key, fix the fresh message and scope, choose r, and form the commitment represented by $A(X)$.
2. Fix the unordered KGO pair pool, every b_i, and r_b, then form the commitment represented by $B(X)$. Only pair orientation remains open.
3. Query H_s, choose r_s, form the honest $S(X)$, and query H_t, the H_u oracles, and H_{agg}.
4. Compute D and orient all 254 pairs from its bits. This changes no input to a base oracle.
5. Expand Equation (8). Because its sole unavailable coefficient is zero, form the SMA proof from public SRS powers.
6. Compute P_h, P_u, P_t from the final ring and produce the later KZG commitments and quotient opening honestly for that ring.
7. Use $sk_\star$ and the known commitment randomness with the honest ccGro16 and CPlink provers. These components are not forged.

If $\delta_\phi = 0$, which occurs with probability $1/q$ in the random-oracle model, discard the transcript and begin another. The retained deterministic row has $\delta_\phi \neq 0$.

5 Correctness and signing consequence

Theorem 1. *Under correctness of ccGro16 and CPlink, the algorithm above produces a paper-level revision-2 LRS signature on an attacker-chosen fresh message for a ring of 1,024 honest KGO outputs, although the attacker's public-key hash is not in the ring. It makes no signing-oracle query and succeeds except for the* $1/q$ *event* $\delta_\phi = 0$.

Proof. Every ring entry is an unchanged honest KGO output; the algorithm only permutes entries inside fixed pairs. Equation (16) makes the coefficient in Equation (10) zero, so every nonzero coefficient of F is supported by a published SRS power and Equation 15 holds. The later KZG commitments and opening are computed from the final ring, so Equation 17 holds by polynomial-commitment correctness. The outside valid secret satisfies the ccGro16 public-key/link-tag relation, and the same committed $\phi_\star$ with known randomness satisfies the CPlink common-opening relation. Correctness supplies accepting proofs for those honest statements. Their conjunction therefore satisfies the paper verifier. The ring pool was chosen disjoint from $\phi_\star$, so the accepted membership statement is false. The message was fixed without a signing-oracle query and is fresh. □

This meets the paper's unforgeability-game boundary: the adversary chooses an ad-hoc ring from honest KGO outputs, corrupts no ring key, requests no signature, and outputs an accepted signature under a ring for which it knows no signing key. Owning a valid key outside that ring does not authorize the signature. Algebraically, the same construction can also produce two accepted signatures at one scope from two outside secrets with distinct link tags, yielding the paper's scoped-linkability consequence. That two-signature extension was not separately executed; the experiment executed one fresh false-membership forgery, which is the primary classification.

6 Cost and full-parameter instantiation

Beyond ordinary proof generation, the attack performs $O(n)$ field operations, 254 public inversions (which may be batch-inverted), a 254-bit decomposition, and a permutation of 1,024 public ring entries. It performs no exponential search and no Fiat–Shamir grinding. The malicious SMA proof uses the same public-SRS multiexponentiation scale as honest full-profile proof generation.

Item	Frozen value
Field	BN254 scalar field in Equation (1)
Public-key mapping	BabyJubJub followed by two-input MultiMiMC7, 91 rounds
Ring/profile	$n = m = 1024$
Pair controls	254 pairs, 508 honest KGO keys; 516 padding positions
Nonzero b_i	Exactly 254, all on honest uncorrupted KGO outputs
Signing queries	0
Setup knowledge	Public SRS only; no powers-of-tau trapdoor
Forged component	Optimized SMA membership statement
Honest components	ccGro16 public-key/link relation and CPlink common opening

Table 1: Full paper profile and attacker boundary.

7 Reproduction and results

The experiment was prospectively sealed before execution. Its seed, message, scope, KGO derivation, pair allocation, exceptional-event rule, and controls were fixed. The implementation uses Python's standard library. It recomputes all 1,024 KGO outputs using the pinned author-source MiMC constants and models SMA/KZG group elements as formal public-SRS coefficient vectors. Exact coefficient identities are stronger than equality at one hidden setup point for the positive row, while making the missing power explicit.

The host reported an AMD EPYC Processor under KVM with 64 virtual CPUs available. Each recorded replay was deliberately fixed to one Python worker and one thread; the available virtual-CPU count is not a worker count.

An independent execution ran the frozen program normally and again under `python -O`; the second run ensures success does not depend on removed assertions. The full replay regenerated all 1,024 public points from the prospectively fixed scalar stream and recomputed the complete transcript. A separately written validator does not import the attack program. It checks the evidence schema, every public position, all BabyJubJub curve equations, all 1,024 MiMC91 key hashes, public-key distinctness, coefficient formulas, residue bits, attacker-key absence, and positive/control predicate records. By itself, that validator does not derive the public points from committed KGO scalars, validate the attacker point, or independently recompute the complete verifier transcript; those checks come from source inspection and the full deterministic rerun.

Run	Wall time	Peak RSS	Workers	Verdict
Ordinary independent replay	63.85 s	24,576 KiB	1	PASS
`python -O` hostile replay	64.27 s	28,220 KiB	1	PASS
Independent evidence validator	–	–	1	PASS for both

Table 2: Sealed full-row deterministic Python paper-model reproduction. The times are not end-to-end author-implementation signing or verification costs.

Both runs emitted the same evidence bytes, with SHA-256

`465e1c4273312f9a4937d3d08b12ca9e09e2117aebe1e8e4d5146a33f939e0a4.`

The evidence records 1,024 distinct ring hashes, exactly 254 nonzero coefficients, 126 selected swaps in this deterministic row, absence of $\phi_\star$, and an accepted missing coefficient of zero.

7.1 Positive and negative controls

Case	Eq. 15	Eq. 17	Composition	Interpretation
Adaptive final ring	PASS	PASS	PASS	Missing coefficient is zero
Fixed pre-challenge order	FAIL	PASS	FAIL	Nonzero unavailable coefficient
Flip pair 0 after proof	PASS	FAIL	FAIL	Later KZG ring binding detects change
Changed message	FAIL	FAIL	FAIL	Message-tagged transcript changes

Table 3: Full-profile checks. The composition column is a paper-level certificate, not a complete byte-level author-verifier execution.

In every row the outside key continues to satisfy the honest ccGro16 relation and the common opening continues to satisfy CPlink. The failures isolate the SMA transcript behavior rather than relying on those two components.

7.2 Exact implementation boundary

The program executes the SMA Equation 15 polynomial identity and the KZG Equation 17 quotient identity at the full paper profile. It checks the exact ccGro16 witness relation and CPlink common opening used by the attacker, but does not instantiate those relations as concrete group proof objects. Its internal `ordinary_accept` flag is therefore a composition certificate under completeness, not an observation from a complete author LRS verifier.

The author repository contains a partial benchmark/reference implementation. No complete external KeyGen/Sign/Verify/Link API, complete ordinary LRS verifier, outer 687-byte codec, or author known-answer test was located in the published artifact. Consequently, no complete author-verifier byte string was available against which to make a stronger claim. The paper semantics remain the normative target.

8 Limitations and repair direction

Ring-order lifecycle. The attack needs freedom to choose the final order of an otherwise fixed multiset after the base commitments. That freedom is present in the paper's ad-hoc-ring API and security game.

An application that externally fixes, canonically orders, and authenticates the ring before all proof commitments could block this exact route; revision 2 does not specify such a lifecycle.

Paper/source scalar mismatch. The paper samples sk from the full BN254 scalar field, while the available cc circuit decomposes 253 bits. The deterministic outside secret used in the experiment is below 2^{253} and so lies in the circuit's domain. This does not resolve the broader mismatch and is another reason not to claim complete source conformance.

Repair. Every Fiat–Shamir oracle that determines the aggregate SMA relation should bind the complete ordered ring statement before any proof value can be adapted. If a deployment instead relies on canonical ordering, the canonical encoding and its authentication must be normative and included in the transcript. Either change requires a new security proof, parameter review, codec definition, and interoperability tests. We do not validate a drop-in patch.

9 Attack history and related technique

No prior public substantive attack against ePrint 2025/1174 revision 2 was located in the searches performed through 2026-10-23. Those searches covered the exact title, identifier, DOI, aliases, authors, repository history, citations, forums, talks, related ring-signature work, and attack terminology. This is a dated, bounded negative result; it cannot exclude private, deleted, unpublished, or unindexed work. Work on other ring signatures does not establish an attack on this target.

The general lesson is classical: a Fiat–Shamir challenge must bind the full statement whose errors it aggregates [2]. Likewise, KZG commitments and missing-power techniques are established tools [3]. We make no claim that adaptive statement selection, binary residue control, or polynomial commitment algebra is new in general. The result identifies their concrete interaction with the revision-2 Constant-Size LRS transcript.

10 Public reproduction package

From `attacks/constant-size-lrs/`, the quick retained-evidence check is

```
python3 -B reproduce.py --reference
python3 -B reproduce-all-rows.py --reference
```

and the fresh full-row replay is

```
python3 -B reproduce.py
./reproducer/reproduce.sh --full
```

The package uses no third-party Python dependency. It pins the target PDF, author-source commit, upstream MiMC source hash, public evidence, direct entrypoints, expected reference outputs, and the independent validator. Raw public evidence reveals no ring secret.

11 Review state

The paper/profile/algebra/security-model review, fresh-context independent reproduction, separate adversarial-result review with all ten binding qualifications preserved, final exact-target attack-history recheck, and independent AI publication-package review passed. This result is classified as `VERIFIED_ATTACK`. Independent human verification has not been performed; it is optional and is not a gate or requirement.

Acknowledgements

The computations were performed on the Norwegian Research and Education Cloud (NREC), using resources provided by the University of Bergen and the University of Oslo.

References

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